MATHEMATICS FOR EVERYONE

For Mathematics
and
For all my fellow Mathematicians

Contents

00. The World of Mathematics **1**

01. Reading Mathematics **3**

02. Drawing Mathematics **5**

03. Writing Mathematics **7**
01. Tips for Using Symbols in Mathematics . . . 7
02. Tips for Using Expressions in Mathematics . . . 8
03. Tips for Using Words and Phrases in Mathematics . . . 9

04. Logic **11**
01. Syntax and Semantics . . . 11
02. Statement or Logical Proposition . . . 11
03. Universal Statement . . . 11
04. Conditional Statement . . . 11
05. Existential Statement . . . 11
06. Universal Existential Statement . . . 11
07. Existential Universal Statement . . . 11
08. Syntax for Propositional Logic . . . 12
09. Well-Formed Formulas . . . 12
10. Semantics for Propositional Logic . . . 12
11. Truth Tables for the Basic Compound WFFs . . . 13
12. Tautology, Contradiction and Contingency . . . 13
13. Logical Implication and Equivalence . . . 14
14. Limitations of Propositional Logic . . . 14
15. Predicate Logic . . . 14
16. Connectives and Predicate Logic . . . 15
17. Quantifiers . . . 15
18. Rules of Inference . . . 16
19. Order of Quantifiers . . . 17
20. Argument . . . 17
21. Conclusion and Premises . . . 17

05. Sets **19**
01. Sets . . . 19
02. Subsets . . . 19
03. Empty Set . . . 19
04. Universal Set . . . 19
05. Proper Subset . . . 19
06. Union of Sets . . . 19
07. Intersection of Sets . . . 19
08. Set Difference . . . 19
09. Complement of Set . . . 19
10. Equality of Sets . . . 19
11. Symmetric Difference of Sets . . . 19
12. Set Identities . . . 19
13. Indexed Collection of Sets . . . 19
14. Union of an Indexed Collection of Sets . . . 19

15. Intersection of an Indexed Collection of Sets . . . 19
16. Disjoint Set . . . 20
17. Partition of Sets . . . 20
18. Power Sets . . . 20
19. Russell's Paradox . . . 20
20. Halting Problem . . . 20

06. Relations **21**
01. Ordered n-tuple . . . 21
02. Cartesian Product of ordered n-tuple . . . 21
03. n-ary Relation . . . 22
04. Binary Relations . . . 22
05. Reflexive Relation . . . 22
06. Symmetric Relation . . . 22
07. Antisymmetric Relation . . . 22
08. Transitive Relation . . . 23
09. Transitive Closure of a Relation . . . 23
10. Equivalence Relations . . . 23
11. Equivalence Class . . . 23
12. Representative of the Class . . . 24
13. Quotient Sets . . . 24
14. Partitions . . . 24
15. Partial Order Relations . . . 25
16. Partially Ordered Set . . . 25
17. Comparable and Noncomparable . . . 25
18. Compatibility of Relation . . . 26
19. Total Order Relation . . . 26
20. Totally Ordered Sets . . . 26
21. Chain of Set . . . 26
22. Maximal Element of Set . . . 26
23. Greatest Element of Set . . . 26
24. Minimal Element of Set . . . 27
25. Least Element of Set . . . 27

07. Functions **29**
01. Functions . . . 29
02. Domain, Codomain, and Range . . . 29
03. Indicator Function . . . 29
04. Step Function . . . 29
05. Greatest Integer Function . . . 29
06. Floor Function . . . 29
07. Image and Preimage . . . 29
08. Injective Function . . . 29
09. Surjective Function . . . 29
10. Bijective Function . . . 29
11. Inverse of Function . . . 29
12. Composition of Function . . . 29
13. Cardinality . . . 29
14. Countable Sets . . . 29
15. The Cantor–Schröder–Bernstein Theorem . . . 29
16. $\mathbb{Q}$ Is Countable . . . 29

Chapter 00 | The World of Mathematics

Chapter 01 | Reading Mathematics

Chapter 02 | Drawing Mathematics

Chapter 03 | Writing Mathematics

01. Tips for Using Symbols in Mathematics

01. Never start a sentence with a symbol.

$x^2 - 6x + 8 = 0$ has two distinct roots. ✗

The equation $x^2 - 6x + 8 = 0$ has two distinct roots. ✓

02. Separate symbols with words, not punctuation, if possible.

With the exception of a, b is the only root of $(x - a)(x - b) = 0$. ✗

With the exception of a, the number b is the only root of $(x - a)(x - b) = 0$. ✓

03. Except when discussing logic, avoid writing the following symbols in your assignments.

$$\Rightarrow, \quad \forall, \quad \exists, \quad \ni, \quad \text{etc.}$$

04. Be careful about using *i.e.* and *e.g.*

i.e. means "that is" and *e.g.* means "for example". It is better not to use them in your assignments; writing the words out is preferable.

$\sqrt{-1}$ and $\lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n$ are not rational numbers, *that is*, i and e are not rational numbers. ✓

05. Write integers as words when they are used as adjectives, and use numerals for large, complicated numbers and exact mathematical values.

There are three roots. ✓

There are 1,527 roots. ✓

Let $n = 3$. ✓

Let $n =$ three. ✗

The interval is $[0, 3]$. ✓

The interval is $[0, \text{three}]$. ✗

06. Don't mix words and symbols improperly.

Every integer ≥ 2 is a prime or is composite. ✗

Every integer exceeding 1 is prime or composite. ✓

If $n \geq 2$ is an integer, then n is prime or composite. ✓

Since $(x-2)(x-3)=0$, it follows that $x=2$ or 3. ✗

Since $(x-2)(x-3)=0$, it follows that $x=2$ or $x=3$. ✓

07. Avoid using a symbol in the statement of a theorem when it is not needed.

Theorem. Every bijective function f has an inverse. ✗

→ The theorem does not depend on what the function is called, so the symbol f provides no value.

Theorem. Every bijective function has an inverse. ✓

08. Explain the meaning of every symbol that you introduce.

09. Use common symbols properly.

m, n	integers
A, B	sets
P, Q, R	propositions
f, g, h	functions

10. Use consistent symbols, unless there is a reason to the contrary.

If x and y are even integers, then $x=2a$ and $y=2r$ for some integers a and b. ✗

If x and y are even integers, then $x=2a$ and $y=2b$ for some integers a and b. ✓

02. Tips for Using Expressions in Mathematics

01. Write short expressions within the text, and lengthy ones as a display.

If expressions are relatively short, then they should probably be written within the text of the proof or discussion. But if they are lengthy, then write the expression as a "display", on a line of its own.

For example, if we expand $(a+b)^6$, then we obtain $(a+b)^6 = a^6 + 6a^5b + 15a^4b^2 + 20a^3b^3 + 15a^2b^4 + 6ab^5 + b^6$. ✗

For example, if we expand $(a+b)^6$, then we obtain ✓

$$(a+b)^6 = a^6 + 6a^5b + 15a^4b^2 + 20a^3b^3 + 15a^2b^4 + 6ab^5 + b^6.$$

02. Display several expressions that are linked by equal signs or inequality symbols.

Since $n = 2k+1$, it follows that ✓

$$\begin{aligned} n^3 + 3n^2 - n + 4 &= (2k+1)^3 + 3(2k+1)^2 - (2k+1) + 4 \\ &= (8k^3 + 12k^2 + 6k + 1) + (12k^2 + 12k + 3) - (2k+1) + 4 \\ &= 8k^3 + 24k^2 + 16k + 7 \\ &= 2(4k^3 + 12k^2 + 8k + 3) + 1. \end{aligned}$$

03. When breaking a lengthy expression across lines, end each line with an operator.

If we have to write a lengthy expression in a text or paragraph and break it into two or more lines, then we should always end each line with an operator, and not start a line with an operator.

For example, if we expand $(a+b)^4$, then we obtain $(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$. ✓

04. Display and label an expression that you want to refer to later.

If we want to refer to an expression that has already appeared, then this expression should be written as a display and labelled.

Expanding $(a+b)^4$, we obtain ✓

$$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4 \tag{1}$$

Setting $a = b = 1$ in (1) gives $16 = 1 + 4 + 6 + 4 + 1$.

03. Tips for Using Words and Phrases in Mathematics

01. I, we, one, let's

I	reflects ego
we	standard mathematical practice
one	a bit awkward
let's	also good, but less professional

It is also possible to avoid these words altogether.

02. Clearly, obviously, of course, certainly

These and similar words turn the reader off if what is written is not evident to the reader. They can give the impression that the author is putting the reader down.

03. Any, each, every

This statement is true for any integer n. ✗

→ Does this mean that it is true for some integer n, or for every integer n? The word *any* is vague here.

This statement is true for every integer n. ✓

Instead of *any*, always use *each* or *every*.

04. Do not write "Since . . . , then . . . "

Use one of these forms instead: "If . . . , then . . . ", "Since . . . , it follows that . . . " or "Since . . . , we have . . . ".

If n^2 is even, then n is even. ✓

Since n^2 is even, it follows that n is even. ✓

Since n^2 is even, n is even. ✓

Since n^2 is even, then n is even. ✗

→ The same applies to *because*.

05. Therefore, thus, hence, consequently, so, it follows that, this implies that

These words are used to proceed from something we have just learned to something else that can be concluded from it. It is good to introduce some variety into your writing.

06. That, which

Both are correct to use in a statement, but they do different jobs. *That* introduces a restrictive clause; *which* does not introduce a restrictive clause.

07. Use good English.

08. Write complete sentences.

09. End each sentence with a period.

10. Capitalize the first word of each sentence.

11. Capitalize theorems and lemmas when they are numbered.

For example: Theorem 1.15, Lemma 4.11, Result 3.14.

12. Do not hyphenate words that begin with the prefix *non*.

For example: nonempty, nonnegative, nondecreasing, nonzero.

13. Watch out for commonly misspelled words.

commutative	corollary	parallel
complement	lemma	preceding
consistent	theorem	proceed
feasible	occurrence	

Chapter 04 | Logic

01. Syntax and Semantics

Syntax and Semantics

Syntax is the alphabet, together with the rules to combine the symbols of the alphabet into well-formed formulas.
Semantics is the interpretation of well-formed formulas.

So in mathematics we should avoid this type of mistake, and make our expressions both well formed syntactically and meaningful semantically. And that is where logical propositions come in.

02. Statement or Logical Proposition

Logical Proposition

A **logical proposition** is a well-formed statement that is either true or false, but not both.

Being *well formed* is a syntactical constraint, and being *either true or false, but not both* is a semantical constraint.

Now we will be discussing two questions:

1. What are the valid symbols we can use? This is the *alphabet*.
2. How can we combine those valid symbols into well-formed expressions? These are the *rules of formation*.

03. Universal Statement

04. Conditional Statement

05. Existential Statement

06. Universal Existential Statement

07. Existential Universal Statement

08. Syntax for Propositional Logic

The alphabet of propositional logic consists of the following.

1. **Propositional variables:** $P, Q, R, \ldots$

   Each propositional variable stands for a logical proposition (a well-formed statement that is either true or false, but not both).

2. **Logical connectives:** used to build more complex expressions by combining one or more propositional variables.

	Connective	Symbol	Pronunciation
unary	negation	$\neg$	"not"
binary	conjunction	$\wedge$	"and"
	disjunction	$\vee$	"or"
	conditional	$\rightarrow$	"if ..., then ..."
	biconditional	$\leftrightarrow$	"if and only if"

3. **Punctuation:** the parentheses ( ), used to clarify the structure of a more complex expression.

09. Well-Formed Formulas

The rules of formation tell us how to combine the alphabet into well-formed formulas.

Well-Formed Formula (WFF)

The well-formed formulas are defined recursively as follows.

1. Every propositional variable $P, Q, R, \ldots$ is a WFF.
2. If P is a WFF, then so is $(\neg P)$.
3. If P and Q are WFFs, then so are

$$(P \wedge Q), \quad (P \vee Q), \quad (P \rightarrow Q), \quad (P \leftrightarrow Q).$$

4. Nothing else is a WFF.

In $(P \rightarrow Q)$: when P is true, Q must be true.

So far we have only discussed the syntactic aspect of propositional logic, that is, how to construct formulas correctly using the rules of WFFs.

But these rules do not guarantee that WFFs will always make sense semantically, that is, they do not guarantee that a WFF will only be true or false, and not both. For that we have to use the truth table.

10. Semantics for Propositional Logic

Our goal is to evaluate the truth value (T or F) of a WFF. Here is how we do it.

1. We give an interpretation to each atomic proposition. This means we assign a truth value to each propositional variable.
2. We extend the evaluation to the basic compound WFFs. This is the essential step: from the truth values of P and Q we get the truth values of

$$(\neg P), \quad (P \wedge Q), \quad (P \vee Q), \quad (P \rightarrow Q), \quad (P \leftrightarrow Q).$$

11. Truth Tables for the Basic Compound WFFs

P	$(\neg P)$
T	F
F	T

P	Q	$(P \wedge Q)$	$(P \vee Q)$	$(P \to Q)$	$(P \leftrightarrow Q)$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

- "If P, then Q" is false if and only if the antecedent P is true and the consequent Q is false.
- The biconditional is true when P and Q have the same truth value.

Disjunction is inclusive. This means $P \vee Q$ is true when P is true, or Q is true, or both P and Q are true.

The conditional is vacuously true when P is false. $(P \to Q)$ says: if P happens, then Q must happen. If P never happened, then the condition was never triggered, and therefore it cannot contradict Q. Hence $(P \to Q)$ is vacuously true.

The WFF above involves 4 individual propositions, so there are $2^4 = 16$ different ways of assigning truth values to these individual atomic propositions.

Filling in a truth table. A strategy to fill in the values of each atomic proposition in each row: number all the proposition columns from right to left. Then

- in the 1st column, change the T/F value after every row;
- in the 2nd column, change it every 2 rows;
- in the 3rd column, change it every 4 rows;
- in the 4th column, change it every 8 rows.

12. Tautology, Contradiction and Contingency

Tautology, Contradiction and Contingency

A proposition is a
- **tautology** if it is always true, under any interpretation;
- **contradiction** if it is always false, under any interpretation;
- **contingency** if it is true under some interpretation, and false under some other interpretation.

Tautologies and contradictions are usually compound propositions. But in some texts atomic propositions are named as

1. **verum** ($\top$), which is constantly true;
2. **falsum** ($\bot$), which is constantly false.

13. Logical Implication and Equivalence

Logical Implication and Logical Equivalence

Let P and Q be two propositions. We say that
- P **logically implies** Q, and denote it by $P \Rightarrow Q$, when $(P \to Q)$ is a tautology;
- P is **logically equivalent** to Q, and denote it by $P \Leftrightarrow Q$, when $(P \leftrightarrow Q)$ is a tautology.

Sufficient and necessary conditions. If $P \Rightarrow Q$, then we say that

1. P is a **sufficient** condition for Q, and
2. Q is a **necessary** condition for P.

14. Limitations of Propositional Logic

There are some limitations to propositional logic. The main one is the inaccessibility of the internal structure of a statement.

15. Predicate Logic

In propositional logic we do not have access to the internal structure of a given statement. To overcome this problem, we need to split the sentence into more detailed components, to recover the internal structure.

A sentence can be split into two components:

1. a **term** (the subject), and
2. a **predicate**.

For example, in *"Mochi is lovely"*, "Mochi" is the term and "is lovely" is the predicate.

Domain of Discourse, Terms and Predicates

- The **domain of discourse** D is the set of all objects of interest.
- A **term** is a symbol that is either a constant or a variable.
  - A **constant** stands for a specific element in D.
  - A **variable** can range over the elements in D.
- An n**-ary predicate** P is a symbol which becomes a formula when applied to n terms.

In predicate logic, each sentence is considered as the result of applying an n-ary predicate to n terms.

Also, the introduction of variables permits us to extend our study from a special case to the general:

- from a special case to a general case;
- from a single object to multiple objects;
- from a static point of view to a dynamical perspective.

16. Connectives and Predicate Logic

Connectives are used to generate compound formulas from atomic ones.

From the above we can say that connectives can be applied in predicate logic to form compound statements, just as we do with propositional logic.

But with some statements in predicate logic, like

Every cat is lovely,

the approach of using connectives fails.

- If D is small, like $D = \{\text{Chowi}, \text{Mochi}\}$, then the above statement can be constructed as

  $$P(\text{Chowi}) \wedge P(\text{Mochi}),$$

  that is, "Chowi is lovely and Mochi is lovely".
- But it becomes impossible to use connectives if D is the set of all the cats in the world.

And this is where quantifiers come to our rescue.

17. Quantifiers

There are two types of quantifiers.

1. **Universal quantifier** $\forall$: "for any", "for all", "for each", "every", ...
2. **Existential quantifier** $\exists$: "there exists", "for some", "for certain", "at least one", ...

In predicate logic, we can combine quantifiers with predicates to generate meaningful propositions. Suppose we have a domain of discourse D and a (unary) predicate P.

- When we apply the predicate to a constant term, we get a formula which bears a truth value.
- But if we apply the predicate to a variable, then we get a well-formed formula that does not bear any truth value, because of the free variable.
- That is where we use quantifiers: to bind the free variable and form a mathematical proposition.

Universal and Existential Statements

Universal statement: $\forall x\, P(x)$, "for all x, $P(x)$". The proposition is true if, whenever we replace x by any constant in the domain of discourse D, the resulting formula is always true, and it is false otherwise.

Existential statement: $\exists x\, P(x)$, "for some x, $P(x)$". The proposition is true if, when we replace x by some particular constant in D, the resulting formula is true, and it is false otherwise.

Negation of quantified statements.

1. $\neg(\forall x\, P(x)) \Leftrightarrow \exists x\, (\neg P(x))$.

   $\forall x\, P(x)$ is false when there exists one constant for which $P(x)$ is false, which is the same as saying that for some x, $P(x)$ is false.

2. $\neg(\exists x\, P(x)) \Leftrightarrow \forall x\, (\neg P(x))$.

   $\exists x\, P(x)$ is false when, no matter which constant we use, $P(x)$ is always false, which is the same as saying that for all x, "not $P(x)$" is a true statement.

18. Rules of Inference

In predicate logic we have two types of statements.

1. **Instantiated statements**, such as $P(c)$.
   - They are achieved by applying a predicate to a constant in the domain of discourse.
   - Instantiated statements concern only a specified instance in the domain of discourse.
2. **Quantified statements**, such as $\forall x\, P(x)$ and $\exists x\, P(x)$.
   - They involve quantifiers and variables.
   - Each quantified statement is about what will happen as the variable ranges through the domain of discourse.

When the domain of discourse contains infinitely many elements, it is impossible for us to reconstruct a quantified statement using merely instantiated statements and connectives.

When we work on deductions, very often we have to carry out some operations, like adding two numbers $(2+3)$ or rotating a triangle. Most of these operations are applied directly to concrete instances instead of to the whole domain of discourse.

To facilitate these kinds of operations in predicate logic, we have some rules of inference that permit us to move back and forth between these two types of statements:

$$\underset{P(c)}{\text{instantiated statement}} \quad \underset{\text{generalization}}{\overset{\text{instantiation}}{\longleftrightarrow}} \quad \underset{\forall x\, P(x),\ \exists x\, P(x)}{\text{quantified statement}}$$

- **Instantiation** is deriving an instantiated statement from a quantified statement.
- **Generalization** is concluding a generalized statement from an instantiated statement.

1. **Universal instantiation.**

   $$\frac{\forall x\, P(x)}{P(c)}, \qquad \text{for an arbitrary constant } c \text{ in the domain of discourse.}$$

   We have the flexibility to choose any c of our preference.

2. **Existential instantiation.**

   $$\frac{\exists x\, P(x)}{P(c)}, \qquad \text{for some fresh constant } c.$$

   Here we have to use a c from the domain of discourse that has never been shown or used before.

3. **Universal generalization.**

   $$\frac{P(c)}{\forall x\, P(x)}, \qquad c \text{ must be arbitrary.}$$

   c should be a generic element: it should not have any property other than the fact that it is an element in D.

4. **Existential generalization.**

   $$\frac{P(c)}{\exists x\, P(x)}, \qquad c \text{ can be specific.}$$

   Here we can assume extra properties of c, as long as we know it is in D.

19. Order of Quantifiers

In a compound statement that involves more than one quantifier:

1. if the quantifiers are the **same**, the order does not matter. For example,

$$\forall x\,\forall y\,P(x,y) \Leftrightarrow \forall y\,\forall x\,P(x,y) \qquad \text{and} \qquad \exists x\,\exists y\,P(x,y) \Leftrightarrow \exists y\,\exists x\,P(x,y);$$

2. if the quantifiers are **different**, the order does matter. For example,

$$\forall x\,\exists y\,P(x,y) \not\Leftrightarrow \exists y\,\forall x\,P(x,y).$$

In $\forall x\,\exists y\,P(x,y)$, the choice of y can depend on x. In $\exists y\,\forall x\,P(x,y)$, the choice of y should not depend on x.

20. Argument

21. Conclusion and Premises

Chapter 05 | Sets

01. Sets

02. Subsets

03. Empty Set

04. Universal Set

05. Proper Subset

06. Union of Sets

07. Intersection of Sets

08. Set Difference

09. Complement of Set

10. Equality of Sets

Definition: Even Integer

An integer n is called even if there exists an integer k such that $n = 2k$.

11. Symmetric Difference of Sets

12. Set Identities

13. Indexed Collection of Sets

14. Union of an Indexed Collection of Sets

15. Intersection of an Indexed Collection of Sets

16. Disjoint Set

17. Partition of Sets

18. Power Sets

19. Russell's Paradox

20. Halting Problem

Chapter 06 | Relations

01. Ordered n-tuple

So far, we have considered sets in which the order and the multiplicity of the element does not matter. For example, the sets $\{a, b\}$, $\{b, a\}$, and $\{a, a, b, b\}$ all represent the same set containing a and b.

But what if the we wanted to consider the collections where the order and multiplicity of the element also does matter.

To study such collections, we introduce the concept of **ordered n-tuples**.

Ordered n-tuple

Let n be a positive integer and let $x_1, x_2, \ldots, x_n$ be (not necessarily distinct) elements. The **ordered n-tuple**, $(x_1, x_2, \ldots, x_n)$, consists of $x_1, x_2, \ldots, x_n$ together with the ordering: first x_1, then x_2, and so forth up to x_n. An ordered 2-tuple is called an **ordered pair**, and an ordered 3-tuple is called an **ordered triple**.
Two ordered n-tuples $(x_1, x_2, \ldots, x_n)$ and $(y_1, y_2, \ldots, y_n)$ are **equal** if, and only if, $x_1 = y_1, x_2 = y_2, \ldots, x_n = y_n$. Symbolically:

$$(x_1, x_2, \ldots, x_n) = (y_1, y_2, \ldots, y_n) \iff x_1 = y_1, x_2 = y_2, \ldots, x_n = y_n.$$

In particular,

$$(a, b) = (c, d) \iff a = c \text{ and } b = d.$$

02. Cartesian Product of ordered n-tuple

Cartesian Product

Given sets $A_1, A_2, \ldots, A_n$, the **Cartesian product** of $A_1, A_2, \ldots, A_n$, denoted $\boldsymbol{A_1 \times A_2 \times \cdots \times A_n}$, is the set of all ordered n-tuples $(a_1, a_2, \ldots, a_n)$ where $a_1 \in A_1, a_2 \in A_2, \ldots, a_n \in A_n$. Symbolically:

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \ldots, a_n \in A_n\}.$$

In particular,

$$A_1 \times A_2 = \{(a_1, a_2) \mid a_1 \in A_1 \text{ and } a_2 \in A_2\}$$

is the Cartesian product of A_1 and A_2.

03. n-ary Relation

n-ary Relation

Given sets $A_1, A_2, \ldots, A_n$, an **n-ary relation** R on $A_1 \times A_2 \times \cdots \times A_n$ is a subset of cartesian product of ordered n tuple $A_1 \times A_2 \times \cdots \times A_n$. The special cases of 2-ary, 3-ary, and 4-ary relations are called **binary**, **ternary**, and **quaternary relations**, respectively.

04. Binary Relations

Binary Relation

Let A, B be two sets. A set $R \subseteq A \times B$ is called a **binary relation** from A to B. If $(x, y) \in R$, we say x is **R-related** to y, and we denote it by xRy.

A binary relation R from A to B is nothing more than a subset of the product set $A \times B$.

This definition not only agrees with our intuition of relations in our daily life, but also provides us with a mathematical language that is flexible and rigorous enough to describe relations among objects in two sets.

- $\emptyset$ is always called the **trivial relation**. It means "nobody in A is related to anybody in B".
- $A \times B$ is called the **universal relation**. It means "everyone in A is related to everyone in B".

05. Reflexive Relation

Reflexive Relation

Let R be a relation on a set A. R is **reflexive** if, and only if, for all $x \in A$, xRx.

06. Symmetric Relation

Symmetric Relation

Let R be a relation on a set A. R is **symmetric** if, and only if, for all $x, y \in A$, ***if*** xRy then yRx.

07. Antisymmetric Relation

Antisymmetric Relation

Let R be a relation on a set A. R is **antisymmetric** if, and only if,

$$\text{for all } a \text{ and } b \text{ in } A, \quad \text{if } aRb \text{ and } bRa \text{ then } a = b.$$

08. Transitive Relation

Transitive Relation

Let R be a relation on a set A. R is **transitive** if, and only if, for all $x, y, z \in A$, ***if*** xRy and yRz then xRz.

09. Transitive Closure of a Relation

Transitive Closure

Let A be a set and R a relation on A. The **transitive closure** of R is the relation R^t on A that satisfies the following three properties:

1. R^t is transitive.
2. $R \subseteq R^t$.
3. If S is any other transitive relation that contains R, then $R^t \subseteq S$.

10. Equivalence Relations

Equivalence Relation

Let A be a set and R a relation on A. R is an **equivalence relation** if, and only if, R is reflexive, symmetric, and transitive.

An equivalence relation R is often denoted by $\sim$. An ordered pair $(x, y) \in R$ is written as $x \sim y$ (x is related to y).

11. Equivalence Class

Equivalence Class

Let $\sim$ be an equivalence relation on a set A. For any element $x \in A$, the set

$$[x] := \{y \in A \mid x \sim y\}$$

is called the **equivalence class** represented by x.

$[x]$, in general, is not an element of A, but a subset of A.

Assume that we have a set A equipped with an equivalence relation, and consider the equivalence class represented by x, as well as that represented by y. What can we say about these as subsets of A? The answer lies in the following two propositions.

Proposition 1

Let A be a non-empty set and $\sim$ be an equivalence relation on A. Then

$$\forall x, y \in A, \quad x \sim y \iff [x] = [y].$$

Proof. ($\Rightarrow$) Assume that $x \sim y$. Let $z \in [y]$ be arbitrary, that is, $y \sim z$. By transitivity, it follows that $x \sim z$, so $z \in [x]$. Hence $[y] \subseteq [x]$. Since $y \sim x$ by symmetry, a similar argument shows that $[x] \subseteq [y]$. We conclude that $[x] = [y]$.

($\Leftarrow$) Assume that $[x] = [y]$. By reflexivity, we have $y \in [y]$. Thus $y \in [x]$, which means that $x \sim y$. □

Proposition 2

Let A be a non-empty set and $\sim$ be an equivalence relation on A. Then

$$\forall x, y \in A, \quad x \not\sim y \iff [x] \cap [y] = \emptyset.$$

Proof. ($\Rightarrow$) Assume that $x \not\sim y$. Suppose, to the contrary, that $z \in [x] \cap [y]$, that is, $x \sim z$ and $y \sim z$. By symmetry, $z \sim y$, and by transitivity it follows that $x \sim y$, which is a contradiction. We conclude that $[x] \cap [y] = \emptyset$.

($\Leftarrow$) Assume that $[x] \cap [y] = \emptyset$. Suppose, to the contrary, that $x \sim y$. Then $y \in [x]$. Moreover, $y \in [y]$ by reflexivity. It turns out that $y \in [x] \cap [y]$, which is a contradiction. Thus $x \not\sim y$. □

12. Representative of the Class

Representative of a Class

Suppose R is an equivalence relation on a set A and S is an equivalence class of R. A **representative** of the class S is any element a such that $[a] = S$.

13. Quotient Sets

Quotient Set

Let A be a non-empty set equipped with an equivalence relation $\sim$. The **quotient set** of A by $\sim$, denoted by

$$A/\sim := \{[x] \mid x \in A\},$$

is the set of all equivalence classes of A under $\sim$.

Each element of $A/\sim$ is an equivalence class of A, that is, a subset of A. In other words, $A/\sim \subseteq \mathcal{P}(A)$.

14. Partitions

Partition

Let A be a non-empty set. A set of its subsets

$$\mathcal{P} = \{U_i \in \mathcal{P}(A) \mid i \in I\},$$

where I is an index set, is called a **partition** of A if

1. $\forall i \in I,\ U_i \neq \emptyset$ (non-emptiness)
2. $\forall i, j \in I,\ i \neq j \Rightarrow U_i \cap U_j = \emptyset$ (pairwise disjointness)
3. $\bigcup_{i \in I} U_i = A$ (cover)

This definition simply means that we cut the set A into different components, and each component U_i in this partition is called a **cell** of the partition.

Theorem: Partition by Equivalence Classes

Let A be a non-empty set and $\sim$ be an equivalence relation on A. Then its quotient set $A/\sim$ forms a partition of A.

Proof. Recall that $A/\sim = \{[x] \mid x \in A\}$.

1. *Non-emptiness.* Let $[x]$ be an arbitrary equivalence class, represented by x. Then by reflexivity, $x \in [x]$.
2. *Pairwise disjointness.* Let $[x], [y]$ be two arbitrary distinct equivalence classes, represented by x and y respectively. Since $[x] \neq [y]$, Proposition 1 gives $x \not\sim y$, and then Proposition 2 gives $[x] \cap [y] = \emptyset$.
3. *Cover.* We prove $\bigcup_{x \in A}[x] = A$ by double set inclusion.
   - $A \subseteq \bigcup_{x \in A}[x]$: let $x \in A$ be arbitrary. By reflexivity, $x \in [x]$, so x is in the union.
   - $\bigcup_{x \in A}[x] \subseteq A$: since $[x] \subseteq A$ for every $x \in A$, so is their union.

□

It is interesting to note that if we were given a partition of the set A in the first place, we could then construct an equivalence relation. If $\mathcal{P} = \{U_i \in \mathcal{P}(A) \mid i \in I\}$ is a partition of A, then the binary relation

$$\forall x, y \in A, \quad x \sim y \;:=\; \exists i \in I,\; x, y \in U_i$$

is an equivalence relation on A.

15. Partial Order Relations

Partial orders are another kind of binary relation.

Partial Order Relation

Let R be a relation defined on a set A. R is a **partial order relation** if, and only if, R is reflexive, antisymmetric, and transitive.

A partial order R is often denoted by $\preceq$. An ordered pair $(x, y) \in R$ is written as $x \preceq y$: x precedes or is equal to y.

16. Partially Ordered Set

Partially Ordered Set

A set A is called a **partially ordered set** (or **poset**) with respect to a relation $\preceq$ if, and only if, $\preceq$ is a partial order relation on A.

17. Comparable and Noncomparable

Comparable and Noncomparable

Suppose $\preceq$ is a partial order relation on a set A. Elements a and b of A are said to be **comparable** if, and only if, either $a \preceq b$ or $b \preceq a$. Otherwise, a and b are called **noncomparable**.

18. Compatibility of Relation

Compatibility of Relations

Given partial order relations $\preceq$ and $\preceq'$ on a set A, $\preceq'$ is **compatible** with $\preceq$ if, and only if, for all a and b in A, if $a \preceq b$ then $a \preceq' b$.

19. Total Order Relation

Total Order Relation

If R is a partial order relation on a set A, and for any two elements a and b in A either aRb or bRa, then R is a **total order relation** on A.

20. Totally Ordered Sets

Totally Ordered Set

A set A is called a **totally ordered set** with respect to a relation $\preceq$ if, and only if, A is partially ordered with respect to $\preceq$ and $\preceq$ is a total order.

21. Chain of Set

Chain

Let A be a set that is partially ordered with respect to a relation $\preceq$. A subset B of A is called a **chain** if, and only if, the elements in each pair of elements in B is comparable. In other words, $a \preceq b$ or $b \preceq a$ for all a and b in B. The **length of a chain** is one less than the number of elements in the chain.

22. Maximal Element of Set

Maximal Element

Let a set A be partially ordered with respect to a relation $\preceq$. An element a in A is called a **maximal element of A** if, and only if, for all b in A, either $b \preceq a$ or b and a are not comparable.

23. Greatest Element of Set

Greatest Element

Let a set A be partially ordered with respect to a relation $\preceq$. An element a in A is called a **greatest element of A** if, and only if, for all b in A, $b \preceq a$.

24. Minimal Element of Set

Minimal Element

Let a set A be partially ordered with respect to a relation $\preceq$. An element a in A is called a **minimal element of A** if, and only if, for all b in A, either $a \preceq b$ or b and a are not comparable.

25. Least Element of Set

Least Element

Let a set A be partially ordered with respect to a relation $\preceq$. An element a in A is called a **least element of A** if, and only if, for all b in A, $a \preceq b$.

Chapter 07 | Functions

01. Functions

02. Domain, Codomain, and Range

03. Indicator Function

04. Step Function

05. Greatest Integer Function

06. Floor Function

07. Image and Preimage

08. Injective Function

09. Surjective Function

10. Bijective Function

11. Inverse of Function

12. Composition of Function

13. Cardinality

14. Countable Sets

15. The Cantor–Schröder–Bernstein Theorem

16. $\mathbb{Q}$ Is Countable